Nunavath Akash

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.08
PM SHRI SCHOOL J N V BAGALUR, BANGALORE URBAN, KARNATAKA	Class XII (CBSE)	2024	91.20%
JAWAHAR NAVODAYA VIDYALAYA MAMNOOR, WARANGAL, TELANGANA	Class X (CBSE)	2022	93.40%

RELEVANT COURSES

- **Computer Science** : Software Development and Fundamentals , Operating Systems , Computer Architecture , Introduction to Programming , Database Management And Systems , Data Structures And Algorithms , Discrete Mathematics , Theory of Computation , Introduction to Compilers.
- **Mathematics & others** : Linear Algebra , Calculus , Probability and Random Variables , Transform Techniques , Introduction to Statistics , Introduction to Artificial Intelligence , Digital Fabrication , Communication Skills.

SKILLS

Languages: Java, C, RISC-V Assembly

Backend & Web: Spring Boot, Spring MVC, Spring Security, JWT, RESTful API design, JSON

Messaging : Apache Kafka

Testing : Junit 5, Spring Boot Testing

Databases : MySQL , MongoDB

Operating Systems : Windows , Ubuntu

Tools & Platforms: Git, GitHub, Postman, IntelliJ IDEA, VS Code, Linux/Ubuntu

PROJECTS

System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ) Scheduler Github

- Redesigned the default Round Robin scheduler in the xv6 operating system by implementing a **4-level System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ)** scheduler.
- Implemented **priority-based process scheduling**, time-slice management, process promotion/demotion, and periodic global priority boosting to improve fairness and prevent starvation.
- Extended the xv6 kernel by adding scheduling metadata to process structures and implementing custom system calls to retrieve process scheduling statistics.
- Developed user-space test programs to evaluate CPU-bound and interactive workloads, validating scheduler behavior and performance under different execution scenarios.

Kafka-Based Order Processing System -- *Java 21, Spring Boot, Apache Kafka, MySQL, MongoDB, Docker* Github

- **Designed** an event-driven microservices system of three independently deployable Spring Boot services (Order, Inventory, Notification) communicating asynchronously through Apache Kafka instead of direct REST calls.
- **Implemented** Kafka producers/consumers to propagate order-lifecycle events (orders.created, inventory.status), driving real-time stock validation and automatic order-status transitions (CREATED CONFIRMED/FAILED).
- **Designed** each service to use the database that fit it best -- MySQL/Spring Data JPA for the Order Service and

MongoDB/Spring Data MongoDB for Inventory and Notification -- with stock-check logic to prevent overselling.
- **Containerized** the full infrastructure (Kafka in KRaft mode, MySQL, MongoDB, Kafka UI) with Docker Compose and exposed documented REST APIs (Swagger UI) for orders, stock, and notifications.

Train Ticket Booking System -- *Java 21, Spring Boot, MongoDB, Spring Security, JWT* Github

- **Built** a secure, layered REST backend (Controller-Service-Repository) with MongoDB persistence across 15+ endpoints for authentication, user, train, and booking operations.

- **Implemented** stateless JWT authentication (JwtFilter, JwtService, CustomUserDetailsService) with BCrypt password hashing and role-based access control distinguishing USER and ADMIN permissions.

- **Designed** multi-passenger ticket booking with seat allocation/release logic, PNR generation, and real-time seat-availability checks backed by MongoDB.

- **Integrated** Spring Mail for automated booking confirmation emails and applied the Strategy pattern (Payment interface with UpiPayment/CardPayment) for pluggable payment handling.

- **Centralized** error handling via a global exception handler covering 5+ domain-specific failures, with credentials externalized through environment variables.

Arbitrary Precision Arithmetic Library -- *Java* Github

- **Engineered** a custom big-number arithmetic library from scratch, implementing addition, subtraction, multiplication, and division on arbitrary-length integers without Java's built-in BigInteger/BigDecimal.
- **Designed** a string-based number representation to overcome primitive data type limits, then packaged the library as a standalone reusable JAR for integration into other Java projects

ADDITIONAL INFORMATION

- Dakshana Scholar (2022-2024) -- selected via NDST for a two-year residential JEE coaching program
- Member, National Sports Organization (NSO)